

RAS PB4.ANL

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*
*          STAAD.Pro
*      Version 2007   Build 02
*      Proprietary Program of
*      Research Engineers, Intl.
*      Date=   MAY 12, 2009
*      Time=   22:12: 9
*
*      USER ID: WS Atkins and Partners Overseas
*****
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INPUT FILE: RAS PB4.STD

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1. STAAD SPACE PAINT BOOTH DUCT SUPPORT
2. START JOB INFORMATION
3. JOB NAME - DUBAI METRO PROJECT
4. JOB CLIENT - GOVERNMENT OF DUBAI _ ROADS & TRANSPORT AUTHORITY
5. ENGINEER DATE 12-MAY-09
6. END JOB INFORMATION
7. INPUT WIDTH 79
8. UNIT METER KN
9. JOINT COORDINATES
10. 11 1.425 0 0.57; 12 2.575 0 0.57; 13 2.575 0 1.715; 14 1.425 0 1.715
11. 21 1.425 1.5 0.57; 22 2.575 1.5 0.57; 23 2.575 1.5 1.715; 24 1.425 1.5 1.715
12. 31 1.425 3 0.57; 32 2.575 3 0.57; 33 2.575 3 1.715; 34 1.425 3 1.715
13. 35 1.425 4.5 0.57; 36 2.575 4.5 0.57; 37 2.575 4.5 1.715; 38 1.425 4.5 1.715
14. 39 1.425 6 0.57; 40 2.575 6 0.57; 41 2.575 6 1.715; 42 1.425 6 1.715
15. 43 1.425 7.5 0.57; 44 2.575 7.5 0.57; 45 2.575 7.5 1.715; 46 1.425 7.5 1.715
16. 47 1.425 9 0.57; 48 2.575 9 0.57; 49 2.575 9 1.715; 50 1.425 9 1.715
17. 51 1.425 10.5 0.57; 52 2.575 10.5 0.57; 53 2.575 10.5 1.715
18. 54 1.425 10.5 1.715; 55 1.425 12 0.57; 56 2.575 12 0.57; 57 2.575 12 1.715
19. 58 1.425 12 1.715; 70 0 0 0; 71 0.55 0 0; 72 0.9 0 0; 73 1.2 0 0; 74 1.425 0 0
20. 75 2 0 0; 76 2.575 0 0; 77 2.8 0 0; 78 3.24 0 0; 79 3.67 0 0; 80 4 0 0
21. 81 0 0 0.345; 82 0.55 0 0.345; 83 0.9 0 0.345; 84 1.2 0 0.345
22. 85 1.425 0 0.345; 86 2 0 0.345; 87 2.575 0 0.345; 88 2.8 0 0.345
23. 89 3.24 0 0.345; 90 3.67 0 0.345; 91 4 0 0.345; 92 0 0 0.57; 93 0.55 0 0.57
24. 94 0.9 0 0.57; 95 1.2 0 0.57; 97 2 0 0.57; 99 2.8 0 0.57; 100 3.24 0 0.57
25. 101 3.67 0 0.57; 102 4 0 0.57; 103 0 0 1.145; 104 0.55 0 1.145
26. 105 0.9 0 1.145; 106 1.2 0 1.145; 108 2.8 0 1.145; 109 3.24 0 1.145
27. 110 3.67 0 1.145; 111 4 0 1.145; 112 0 0 1.715; 113 0.55 0 1.715
28. 114 0.9 0 1.715; 115 1.2 0 1.715; 117 2.8 0 1.715; 118 3.24 0 1.715
29. 119 3.67 0 1.715; 120 4 0 1.715; 121 0 0 1.95; 122 0.55 0 1.95; 123 0.9 0 1.95
30. 124 1.2 0 1.95; 125 1.425 0 1.95; 126 2 0 1.95; 127 2.575 0 1.95
31. 128 2.8 0 1.95; 129 3.24 0 1.95; 130 3.67 0 1.95; 131 4 0 1.95; 132 0 0 2.1
32. 133 0.55 0 2.1; 134 0.9 0 2.1; 135 1.2 0 2.1; 136 1.425 0 2.1; 137 2 0 2.1
33. 138 2.575 0 2.1; 139 2.8 0 2.1; 140 3.24 0 2.1; 141 3.67 0 2.1; 142 4 0 2.1
34. 144 0.55 0 2.4; 145 0.9 0 2.4; 146 1.2 0 2.4; 147 1.425 0 2.4; 148 2 0 2.4
35. 149 2.575 0 2.4; 150 2.8 0 2.4; 151 3.24 0 2.4; 152 3.67 0 2.4; 153 4 0 2.4
36. 154 2.575 0 1.145; 155 2 0 1.145; 156 1.425 0 1.145; 157 2 0 1.715
37. 158 0.55 0 2.7; 160 0.9 0 2.7; 161 1.2 0 2.7; 162 1.425 0 2.7; 163 2 0 2.7
38. 164 2.575 0 2.7; 165 2.8 0 2.7; 166 3.24 0 2.7; 167 3.67 0 2.7; 168 4 0 2.7
39. MEMBER INCIDENCES
40. 21 21 22; 22 22 23; 23 23 24; 24 24 21; 31 31 32; 32 32 33; 33 33 34; 34 34 31
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41. 1001 11 21; 1002 12 22; 1003 13 23; 1004 14 24; 1101 24 11; 1102 24 31
42. 1201 22 13; 1202 22 33; 1301 23 14; 1302 23 34; 1401 21 12; 1402 21 32
43. 2001 21 31; 2002 22 32; 2003 23 33; 2004 24 34; 2005 35 36; 2006 36 37
44. 2007 37 38; 2008 38 35; 2009 39 40; 2010 40 41; 2011 41 42; 2012 42 39
45. 2013 31 35; 2014 32 36; 2015 33 37; 2016 34 38; 2017 38 31; 2018 38 39
46. 2019 36 33; 2020 36 41; 2021 37 34; 2022 37 42; 2023 35 32; 2024 35 40
47. 2025 35 39; 2026 36 40; 2027 37 41; 2028 38 42; 2029 43 44; 2030 44 45
48. 2031 45 46; 2032 46 43; 2033 47 48; 2034 48 49; 2035 49 50; 2036 50 47
49. 2037 39 43; 2038 40 44; 2039 41 45; 2040 42 46; 2041 46 39; 2042 46 47
50. 2043 44 41; 2044 44 49; 2045 45 42; 2046 45 50; 2047 43 40; 2048 43 48
51. 2049 43 47; 2050 44 48; 2051 45 49; 2052 46 50; 2053 51 52; 2054 52 53
52. 2055 53 54; 2056 54 51; 2057 55 56; 2058 56 57; 2059 57 58; 2060 58 55
53. 2061 47 51; 2062 48 52; 2063 49 53; 2064 50 54; 2065 54 47; 2066 54 55
54. 2067 52 49; 2068 52 57; 2069 53 50; 2070 53 58; 2071 51 48; 2072 51 56
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55. 2073 51 55; 2074 52 56; 2075 53 57; 2076 54 58
56. ELEMENT INCIDENCES SHELL
57. 2087 70 71 82 81; 2088 71 72 83 82; 2089 72 73 84 83; 2090 73 74 85 84
58. 2091 74 75 86 85; 2092 75 76 87 86; 2093 76 77 88 87; 2094 77 78 89 88
59. 2095 78 79 90 89; 2096 79 80 91 90; 2097 81 82 93 92; 2098 82 83 94 93
60. 2099 83 84 95 94; 2100 84 85 11 95; 2101 85 86 97 11; 2102 86 87 12 97
61. 2103 87 88 99 12; 2104 88 89 100 99; 2105 89 90 101 100; 2106 90 91 102 101
62. 2107 92 93 104 103; 2108 93 94 105 104; 2109 94 95 106 105
63. 2114 99 100 109 108; 2115 100 101 110 109; 2116 101 102 111 110
64. 2117 103 104 113 112; 2118 104 105 114 113; 2119 105 106 115 114
65. 2124 108 109 118 117; 2125 109 110 119 118; 2126 110 111 120 119
66. 2127 112 113 122 121; 2128 113 114 123 122; 2129 114 115 124 123
67. 2130 115 14 125 124; 2133 13 117 128 127; 2134 117 118 129 128
68. 2135 118 119 130 129; 2136 119 120 131 130; 2137 121 122 133 132
69. 2138 122 123 134 133; 2139 123 124 135 134; 2140 124 125 136 135
70. 2141 125 126 137 136; 2142 126 127 138 137; 2143 127 128 139 138
71. 2144 128 129 140 139; 2145 129 130 141 140; 2146 130 131 142 141
72. 2148 133 134 145 144; 2149 134 135 146 145; 2150 135 136 147 146
73. 2151 136 137 148 147; 2152 137 138 149 148; 2153 138 139 150 149
74. 2154 139 140 151 150; 2155 140 141 152 151; 2156 141 142 153 152
75. 2167 95 11 156 106; 2168 11 97 155 156; 2169 97 12 154 155; 2170 12 99 108 154
76. 2171 106 156 14 115; 2175 156 155 157 14; 2176 155 154 13 157
77. 2177 14 157 126 125; 2178 157 13 127 126; 2179 154 108 117 13
78. 2181 144 145 160 158; 2182 145 146 161 160; 2183 146 147 162 161
79. 2184 147 148 163 162; 2185 148 149 164 163; 2186 149 150 165 164
80. 2187 150 151 166 165; 2188 151 152 167 166; 2189 152 153 168 167
81. ELEMENT PROPERTY
82. 2100 TO 2103 2130 2133 2167 TO 2171 2175 TO 2179 THICKNESS 1.95
83. 2087 TO 2099 2104 TO 2109 2114 TO 2119 2124 TO 2129 2134 TO 2146 2148 TO 2156 -
84. 2181 TO 2189 THICKNESS 0.75
85. MEMBER RELEASE
86. 1001 TO 1004 START MX MY MZ
87. 1101 1102 1201 1202 1301 1302 2017 TO 2022 2041 TO 2046 2065 TO 2069 -
88. 2070 BOTH MX MY MZ
89. 1401 1402 2023 2024 2047 2048 2071 2072 BOTH MX MY MZ
90. MEMBER PROPERTY BRITISH
91. 1001 TO 1004 2001 TO 2004 2013 TO 2016 2025 TO 2028 2037 TO 2040 2049 TO 2052 -
92. 2061 TO 2064 2073 TO 2076 TABLE ST TUB2002006.0
93. 21 TO 24 31 TO 34 1101 1102 1201 1202 1301 1302 1401 1402 2005 TO 2012 2017 -
94. 2018 TO 2024 2029 TO 2036 2041 TO 2048 2053 TO 2060 2065 TO 2071 -
95. 2072 TABLE ST TUBE TH 0.0045 WT 0.125 DT 0.125
96. DEFINE MATERIAL START
PAINT BOOTH DUCT SUPPORT
-- PAGE NO. 3
97. ISOTROPIC STEEL
98. E 2.05E+008
99. POISSON 0.3
100. DENSITY 76.8195
101. ALPHA 1.2E-005
102. DAMP 0.03
103. ISOTROPIC CONCRETE
104. E 2.17185E+007
105. POISSON 0.17
106. DENSITY 23.5616
107. ALPHA 1E-005
108. DAMP 0.05
109. END DEFINE MATERIAL
110. CONSTANTS
111. MATERIAL STEEL MEMB 21 TO 24 31 TO 34 1001 TO 1004 1101 1102 1201 1202 1301 -
112. 1302 1401 1402 2001 TO 2076
113. MATERIAL CONCRETE MEMB 2087 TO 2109 2114 TO 2119 2124 TO 2130 2133 TO 2146 -
114. 2148 TO 2156 2167 TO 2171 2175 TO 2179 2181 TO 2189
115. SUPPORTS
116. 11 TO 14 70 TO 95 97 99 TO 106 108 TO 115 117 TO 142 144 TO 158 160 TO 167 -
117. 168 ELASTIC MAT DIRECT Y SUBGRADE 18000
118. LOAD 1 SELFWEIGHT
119. SELFWEIGHT Y -1
120. LOAD 2 SDL
121. MEMBER LOAD
122. 21 23 31 33 2005 2007 2009 2011 2029 2031 2033 2035 2053 2055 2057 -
123. 2059 CON GY -0.35
124. LOAD 11 WL1
125. MEMBER LOAD
126. *WL = 1.054KPA
127. 1001 1002 2001 2002 2013 2014 2025 2026 2037 2038 2049 2050 2061 2062 2073 -

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128. 2074 UNI GZ -0.71
 129. LOAD 12 WL2
 130. MEMBER LOAD
 131. *WL = 1.67KPA
 132. 1001 1004 2001 2004 2013 2016 2025 2028 2037 2040 2049 2052 2061 2064 2073 -
 133. 2076 UNI GX -1.13
 134. LOAD 21 EQ1 (EARTHQUAKE LOAD)
 135. JOINT LOAD
 136. 21 22 31 32 35 36 39 40 43 44 47 48 51 52 55 56 FZ -0.3
 137. 23 24 33 34 37 38 41 42 45 46 49 50 53 54 57 58 FZ -0.3
 138. LOAD 22 EQ2 (EARTHQUAKE LOAD)
 139. JOINT LOAD
 140. 21 24 31 34 35 38 39 42 43 46 47 50 51 54 55 58 FX -0.3
 141. 22 23 32 33 36 37 40 41 44 45 48 49 52 53 56 57 FX -0.3
 142. LOAD COMB 1001 1.4DL + 1.4SDL
 143. 1 1.4 2 1.4
 144. LOAD COMB 1011 1.2DL + 1.2SDL + 1.2WL1
 145. 1 1.2 2 1.2 11 1.2
 146. LOAD COMB 1111 1.4DL + 1.4WL1
 147. 1 1.4 2 1.4 11 1.4
 148. LOAD COMB 1211 1.0DL + 1.4WL1
 149. 1 1.0 2 1.0 11 1.4
 150. LOAD COMB 1311 1.2DL + 1.2SDL - 1.2WL1
 151. 1 1.2 2 1.2 11 -1.2
 152. LOAD COMB 1411 1.4DL - 1.4WL1
 PAINT BOOTH DUCT SUPPORT

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153. 1 1.4 2 1.4 11 -1.4
 154. LOAD COMB 1511 1.0DL - 1.4WL1
 155. 1 1.0 2 1.0 11 -1.4
 156. LOAD COMB 1012 1.2DL + 1.2SDL + 1.2WL2
 157. 1 1.2 2 1.2 12 1.2
 158. LOAD COMB 1112 1.4DL + 1.4WL2
 159. 1 1.4 2 1.4 12 1.4
 160. LOAD COMB 1212 1.0DL + 1.4WL2
 161. 1 1.0 2 1.0 12 1.4
 162. LOAD COMB 1312 1.2DL + 1.2SDL - 1.2WL2
 163. 1 1.2 2 1.2 12 -1.2
 164. LOAD COMB 1412 1.4DL - 1.4WL2
 165. 1 1.4 2 1.4 12 -1.4
 166. LOAD COMB 1512 1.0DL - 1.4WL2
 167. 1 1.0 2 1.0 12 -1.4
 168. LOAD COMB 1021 1.2DL + 1.2SDL + 1.0EQ1
 169. 1 1.2 2 1.2 21 1.0
 170. LOAD COMB 1121 1.2DL + 1.2SDL - 1.0EQ1
 171. 1 1.2 2 1.2 21 -1.0
 172. LOAD COMB 1221 0.9DL + 1.0EQ1
 173. 1 0.9 2 0.9 21 1.0
 174. LOAD COMB 1321 0.9DL - 1.0EQ1
 175. 1 0.9 2 0.9 21 -1.0
 176. LOAD COMB 1022 1.2DL + 1.2SDL + 1.0EQ2
 177. 1 1.2 2 1.2 22 1.0
 178. LOAD COMB 1122 1.2DL + 1.2SDL - 1.0EQ2
 179. 1 1.2 2 1.2 22 -1.0
 180. LOAD COMB 1222 0.9DL + 1.0EQ2
 181. 1 0.9 2 0.9 22 1.0
 182. LOAD COMB 1322 0.9DL - 1.0EQ2
 183. 1 0.9 2 0.9 22 -1.0
 184. *SLS
 185. LOAD COMB 2001 1.0DL + 1.0SDL
 186. 1 1.0 2 1.0
 187. LOAD COMB 2011 DL + SDL + WL1
 188. 1 1.0 2 1.0 11 1.0
 189. LOAD COMB 2111 DL + SDL - WL1
 190. 1 1.0 2 1.0 11 -1.0
 191. LOAD COMB 2012 DL + SDL + WL2
 192. 1 1.0 2 1.0 12 1.0
 193. LOAD COMB 2112 DL + SDL - WL2
 194. 1 1.0 2 1.0 12 -1.0
 195. *
 196. PERFORM ANALYSIS
 PAINT BOOTH DUCT SUPPORT

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P R O B L E M S T A T I S T I C S

NUMBER OF JOINTS/MEMBER+ELEMENTS/SUPPORTS = 129/ 174/ 97
 ORIGINAL/FINAL BAND-WIDTH= 117/ 17/ 108 DOF
 TOTAL PRIMARY LOAD CASES = 6, TOTAL DEGREES OF FREEDOM = 774
 SIZE OF STIFFNESS MATRIX = 84 DOUBLE KILO-WORDS
 REQD/AVAIL. DISK SPACE = 13.5/ 6713.7 MB, EXMEM = 1187.9 MB

197. LOAD LIST 2001 2011 2012 2111 2112
 198. PRINT JOINT DISPLACEMENTS LIST 55
 JOINT DISPLACE LIST 55
 PAINT BOOTH DUCT SUPPORT

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JOINT DISPLACEMENT (CM RADIANS) STRUCTURE TYPE = SPACE

JOINT	LOAD	X-TRANS	Y-TRANS	Z-TRANS	X-ROTAN	Y-ROTAN	Z-ROTAN
55	2001	0.0118	-0.1754	-0.2348	-0.0002	0.0000	0.0000
	2011	0.0429	-0.2572	-1.5914	-0.0014	0.0000	0.0000
	2111	-0.0192	-0.0937	1.1219	0.0010	0.0000	0.0000
	2012	-1.2785	-0.2413	-0.2005	-0.0002	0.0000	0.0011
	2112	1.3022	-0.1096	-0.2690	-0.0002	0.0000	-0.0012

***** END OF LATEST ANALYSIS RESULT *****

199. PRINT SUPPORT REACTION ALL
 SUPPORT REACTION ALL
 PAINT BOOTH DUCT SUPPORT

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SUPPORT REACTIONS -UNIT KN METE STRUCTURE TYPE = SPACE

JOINT	LOAD	FORCE-X	FORCE-Y	FORCE-Z	MOM-X	MOM-Y	MOM Z
11	2001	0.10	4.88	0.09	0.00	0.00	0.00
	2011	0.47	6.72	8.80	0.00	0.00	0.00
	2111	-0.26	3.04	-8.62	0.00	0.00	0.00
	2012	0.46	5.92	0.66	0.00	0.00	0.00
	2112	-0.25	3.83	-0.48	0.00	0.00	0.00
12	2001	-0.10	4.91	0.09	0.00	0.00	0.00
	2011	-0.48	6.83	0.31	0.00	0.00	0.00
	2111	0.27	3.00	-0.13	0.00	0.00	0.00
	2012	13.09	3.82	-0.49	0.00	0.00	0.00
	2112	-13.29	6.01	0.67	0.00	0.00	0.00
13	2001	-0.10	4.29	-0.09	0.00	0.00	0.00
	2011	0.26	3.37	8.22	0.00	0.00	0.00
	2111	-0.47	5.21	-8.40	0.00	0.00	0.00
	2012	-0.41	3.31	0.51	0.00	0.00	0.00
	2112	0.21	5.28	-0.68	0.00	0.00	0.00
14	2001	0.10	4.26	-0.09	0.00	0.00	0.00
	2011	-0.25	3.26	-0.29	0.00	0.00	0.00
	2111	0.46	5.25	0.11	0.00	0.00	0.00
	2012	13.98	5.42	-0.67	0.00	0.00	0.00
	2112	-13.78	3.09	0.49	0.00	0.00	0.00
70	2001	0.00	1.52	0.00	0.00	0.00	0.00
	2011	0.00	2.45	0.00	0.00	0.00	0.00
	2111	0.00	0.59	0.00	0.00	0.00	0.00
	2012	0.00	2.59	0.00	0.00	0.00	0.00
	2112	0.00	0.46	0.00	0.00	0.00	0.00